

Individual Contribution Report

In our group project, we performed statistical analysis such as Chi-square testing and Logistic Regression to study factors affecting university eligibility.

For my individual application, I extended this work by implementing these statistical methods into a fully functional and interactive Streamlit-based system.

Summary of Technical Implementations

Although Logistic Regression was used in the group analysis, I implemented and integrated the model into the application to function as a real-time eligibility predictor. Specifically, I converted categorical grades into numerical values, split the dataset into training and testing sets, trained a Logistic Regression model for that.

Additionally, I transformed the Chi-Square testing from a manual calculation into an interactive diagnostic tool, allowing users to perform automated hypothesis testing and contingency mapping between any two subjects in the dataset.

Key Contribution: Smart-Rule Hybrid Eligibility Predictor

I developed a Hybrid Prediction Engine to resolve a critical limitation in standard Machine Learning: in a pure probabilistic model like Logistic Regression, high weights from two "A" grades can mathematically mask a "Fail" (F) grade, leading to a "False Positive".

Since the Sri Lankan A/L system mandates absolute disqualification for any "F" grade, I developed a system that integrates Deterministic Logic with Probabilistic Modeling.

Before running the AI model, the system first checks:

- **The Rule-Based Filter (Hard Logic):** Before the data reaches the AI, my custom "Guardrail" code scans the input for any 'F' grade. If detected, the system immediately triggers an "Ineligible" status, bypassing the machine learning inference entirely to ensure legal accuracy.
- **The ML Inference (Soft Logic):** If the student meets the minimum requirement (at least 3 'S' passes), the data proceeds to the Logistic Regression model. Here, the AI analyzes the grade combination against 2020 competitiveness patterns to calculate the statistical probability of university admission. This ensures that predictions follow official Sri Lankan admission regulations and prevents unrealistic AI outputs.

- Prediction page showing automatic “Ineligible” result when an ‘F’ grade is entered.

The screenshot shows the 'University Eligibility Predictor' web application. On the left is a dark blue sidebar with the 'Student Portal' header and a navigation menu. The main content area has a white background. At the top right of the main area is a 'Deploy' button. Below the title is a disclaimer box. The form contains three dropdown menus for 'Subject 1 Grade' (A), 'Subject 2 Grade' (A), and 'Subject 3 Grade' (F). A yellow 'Run Prediction' button is below the form. The result is displayed in a pink box: 'Outcome: INELIGIBLE'. A yellow box below explains the reasoning: 'Reasoning: Sri Lankan university entry rules require a minimum of an 'S' pass in all 3 subjects. An 'F' results in automatic ineligibility.'

Student Portal

Navigation

- Project & Data Explorer
- Exploratory Analysis
- Subject Dependency (Chi-Square)
- Eligibility Model (ML)
- Predict Your Eligibility

University Eligibility Predictor

⚠️ Disclaimer: This predictor is a statistical tool based on 2020 historical data. It is NOT an official admission letter. Official eligibility is verified only by the UGC.

Subject 1 Grade: A Subject 2 Grade: A Subject 3 Grade: F

Run Prediction

Outcome: INELIGIBLE

Reasoning: Sri Lankan university entry rules require a minimum of an 'S' pass in all 3 subjects. An 'F' results in automatic ineligibility.

- Prediction page showing eligibility result with probability when all subjects are passed.

The screenshot shows the 'University Eligibility Predictor' web application with the same sidebar as the first image. The main content area shows the 'Run Prediction' button with 'Subject 1 Grade' (C), 'Subject 2 Grade' (S), and 'Subject 3 Grade' (S) selected. The result is displayed in a green box: 'Outcome: ELIGIBLE'. Below this, a box shows the 'AI Confidence Level' as 97.1%.

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University Eligibility Predictor

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Subject 1 Grade: C Subject 2 Grade: S Subject 3 Grade: S

Run Prediction

Outcome: ELIGIBLE

AI Confidence Level
97.1%

Conclusion

Through this individual work, I transformed the group’s statistical analysis into an interactive decision-support system. I integrated Logistic Regression into a live prediction tool, implemented dynamic Chi-square testing and added a rule-based safeguard to ensure realistic and reliable eligibility decisions.